Project Initialization and Planning Phase

Date	12 July 2024
Team ID	-
Project Title	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The Voice-Based Concept Understanding Analyser (VBCUA) is an AI-powered educational application that evaluates a student's conceptual understanding through spoken explanations. The system converts speech into text using Faster Whisper, compares it with expert reference concepts using Sentence-BERT, calculates semantic similarity, performs keyword matching, generates intelligent feedback, stores results in a SQLite database, and produces downloadable PDF reports. This solution reduces manual evaluation effort while providing fast, accurate, and objective assessment.

Project Overview	
Objective	To develop an AI-powered system that evaluates students' conceptual understanding by analyzing spoken explanations using speech recognition and semantic similarity techniques, providing accurate scores, personalized feedback, and detailed reports.
Scope	The project enables users to upload audio explanations, perform speech-to-text conversion, compare explanations with reference concepts, generate AI-based scores, provide intelligent feedback, store analysis history, visualize performance, and generate downloadable PDF reports.
Problem Statement	

Description	Traditional evaluation methods rely heavily on manual assessment and keyword matching, which are time-consuming and often fail to measure true conceptual understanding. There is a need for an intelligent system capable of evaluating spoken explanations based on semantic meaning rather than exact word matching..
Impact	The proposed solution improves the accuracy, consistency, and efficiency of concept evaluation. It reduces teachers' workload, provides immediate feedback to students, supports concept-based learning, and enhances the overall educational assessment process.
Proposed Solution	
Approach	The system uses Faster Whisper to convert speech into text and Sentence-BERT to measure semantic similarity between the student's explanation and expert reference concepts. Keyword matching, AI-based scoring, and intelligent feedback generation further enhance the assessment process. All results are stored in an SQLite database and can be exported as PDF reports.
Key Features	• Audio Upload • Speech-to-Text using Faster Whisper • Semantic Similarity using Sentence-BERT • Keyword Matching • AI Score Generation • Personalized Feedback • SQLite Database Storage • Dashboard Analytics • PDF Report Generation
Resource Requirements	

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	Computing Device	Intel Core i3/i5 or higher
Memory	RAM specifications	Minimum 4 GB RAM (8 GB Recommended)

Storage	Disk space for data, models, and logs	10 GB Free Disk Space
Software		
Frameworks Streamlit	Python frameworks	Flask
Libraries Faster Whisper Sentence-Transformers Transformers Torch Librosa NumPy Pandas Scikit-learn Plotly Matplotlib ReportLab SoundFile Python-dotenv	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment Visual Studio Code Python Virtual Environment (venv) Git GitHub	IDE	Jupyter Notebook, pycharm
Data		
Data Reference Concepts: • Machine Learning • Artificial Intelligence • Cloud Computing • Data Science • Database Management System • Operating System	Input Data: Student voice recordings (.wav, .mp3, .ogg, .m4a, .webm)	Output Data: Speech transcripts, semantic similarity scores, keyword matching results, AI feedback, dashboard analytics, and PDF reports.